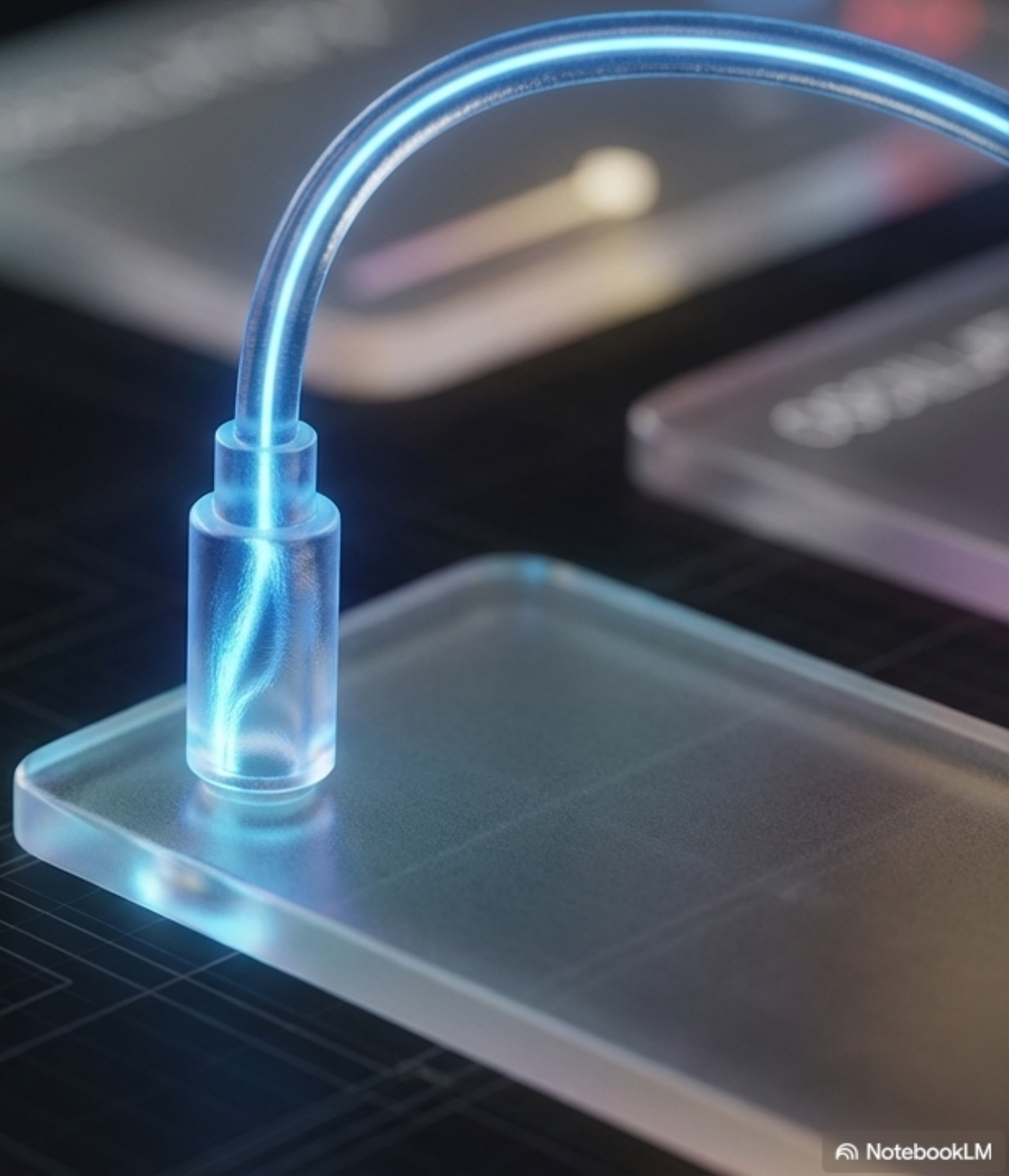


Luminous Architecture

Visualizing the 37-Node Field
Additive Synthesizer



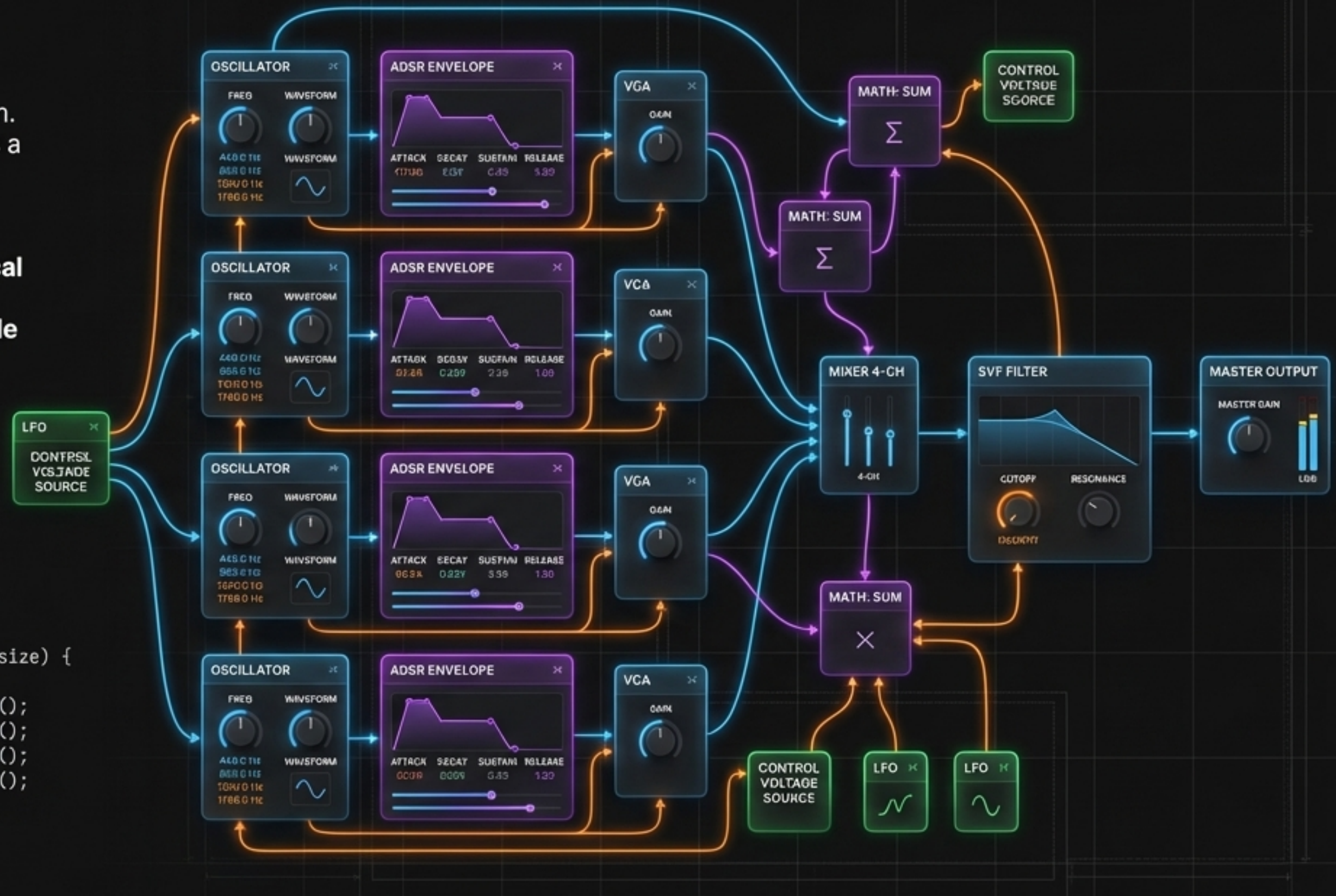
The Cognitive Load of Sound Design

Additive synthesis requires immense precision. Building a 4-partial harmonic stack generates a dense, 37-node web of audio signals, math operators, and control voltages.

The design must transform extreme technical density into a highly readable, cognitively intuitive interface without sacrificing a single routing connection.

```
daisysp::Oscillator osc1, osc2, osc3, osc4;  
daisysp::Adsr env1, env2, env3, env4;  
daisysp::Svf filter;  
float freq1, freq2, freq3, freq4;  
float gain1, gain2, gain3, gain4;  
float masterGain = 0.8f;  
float output;
```

```
void AudioCallback(float *in, float *out, size_t size) {  
    for (size_t i = 0; i < size; i++) {  
        float sig1 = osc1.Process() * env1.Process();  
        float sig2 = osc2.Process() * env2.Process();  
        float sig3 = osc3.Process() * env3.Process();  
        float sig4 = osc4.Process() * env4.Process();  
        float mixed = sig1 + sig2 + sig3 + sig4;  
        filter.Process(mixed);  
        out[i] = filter.Low() * masterGain;  
    }  
}
```



The Studio Onyx Canvas

A #121212 foundation engineered for maximum contrast. The faint blueprint grid provides spatial orientation for complex modular patching while receding silently into the background.

Neon Glassmorphism

- Frosted 2D Glass Panels
- Backdrop-Filter Blur
- Soft Drop Shadows
- Glowing Neon Borders



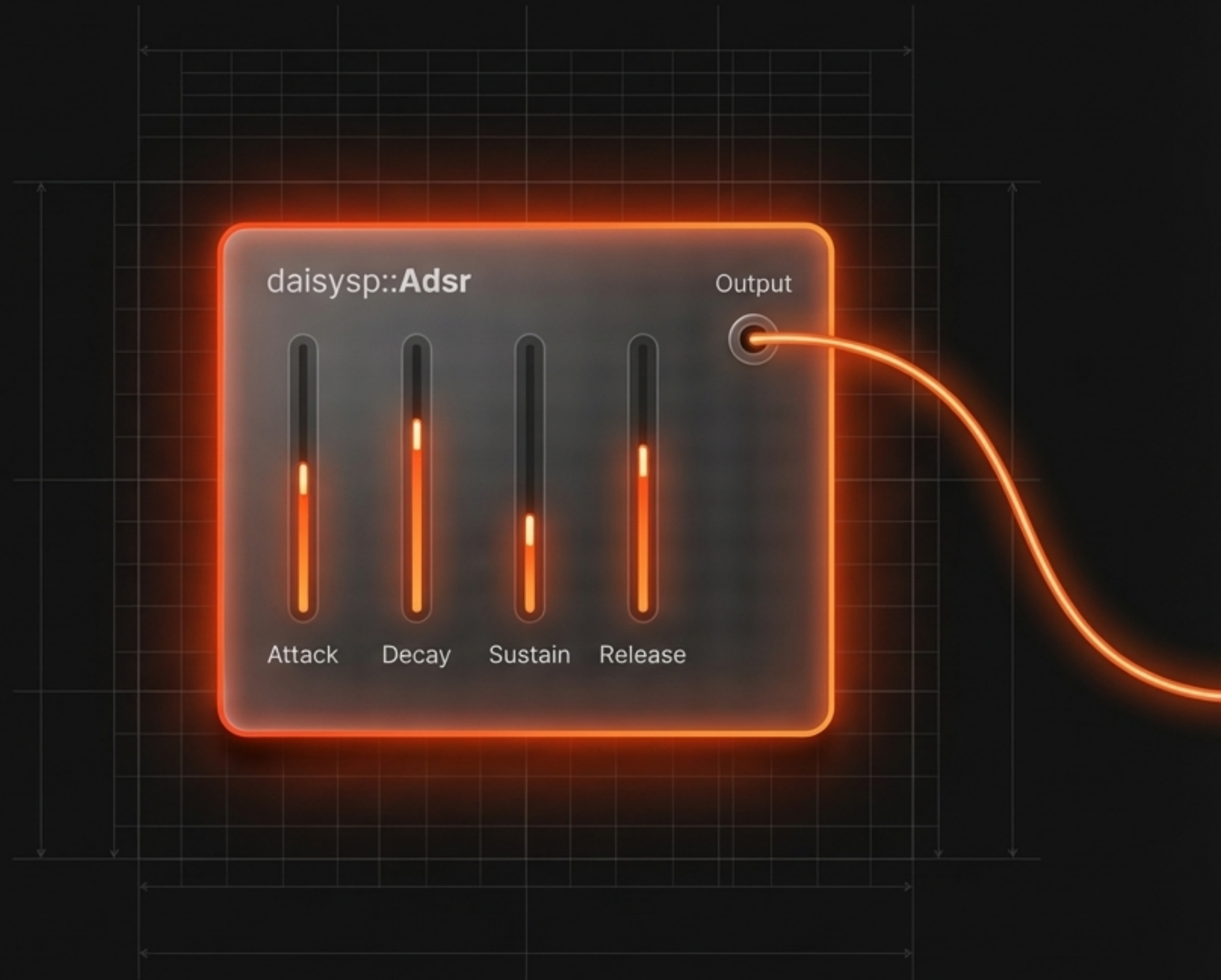
Toxic Green: The Human Element

Strict semantic color coding dictates the visual hierarchy. Toxic Green represents the physical world: hardware input controls, MIDI notes, gates, and manual UI interaction.



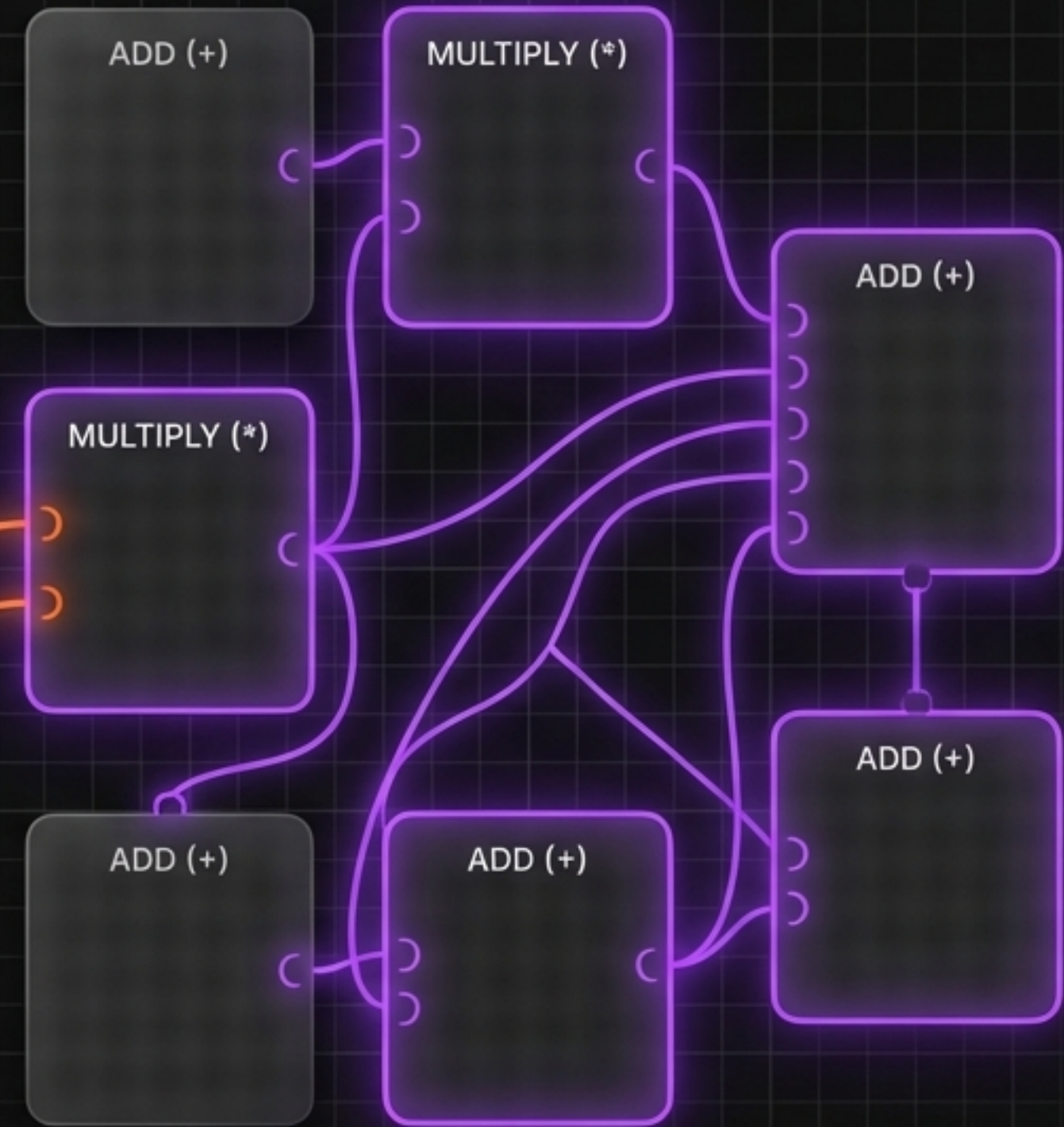
Burning Orange: Control & Modulation

The nervous system of the synthesizer. **Burning Orange** maps all Control Voltage (CV), envelope generators, and Low-Frequency Oscillators (LFOs). These paths dictate how sound breathes and moves over time.



Electric Violet: Signal Logic

The architectural routing brain. Electric Violet highlights the hidden mathematics: harmonic multipliers, summing mixers, and signal logic nodes that construct complex timbres.



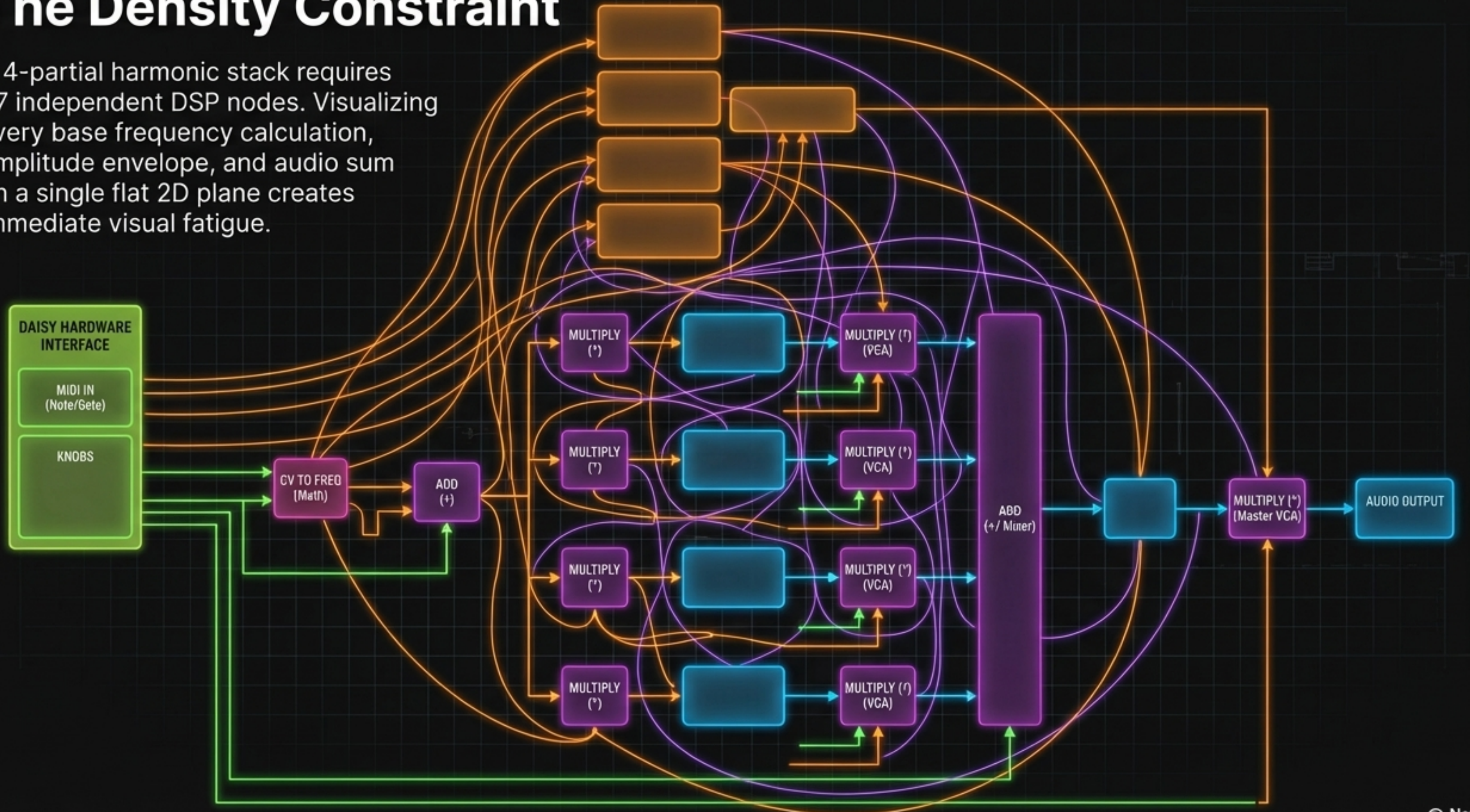
Neon Blue: The Audio Path

The spine of the instrument. Neon Blue is reserved exclusively for the main audio signal path, carving a clear, unambiguous visual route from the raw oscillator to the final master output.



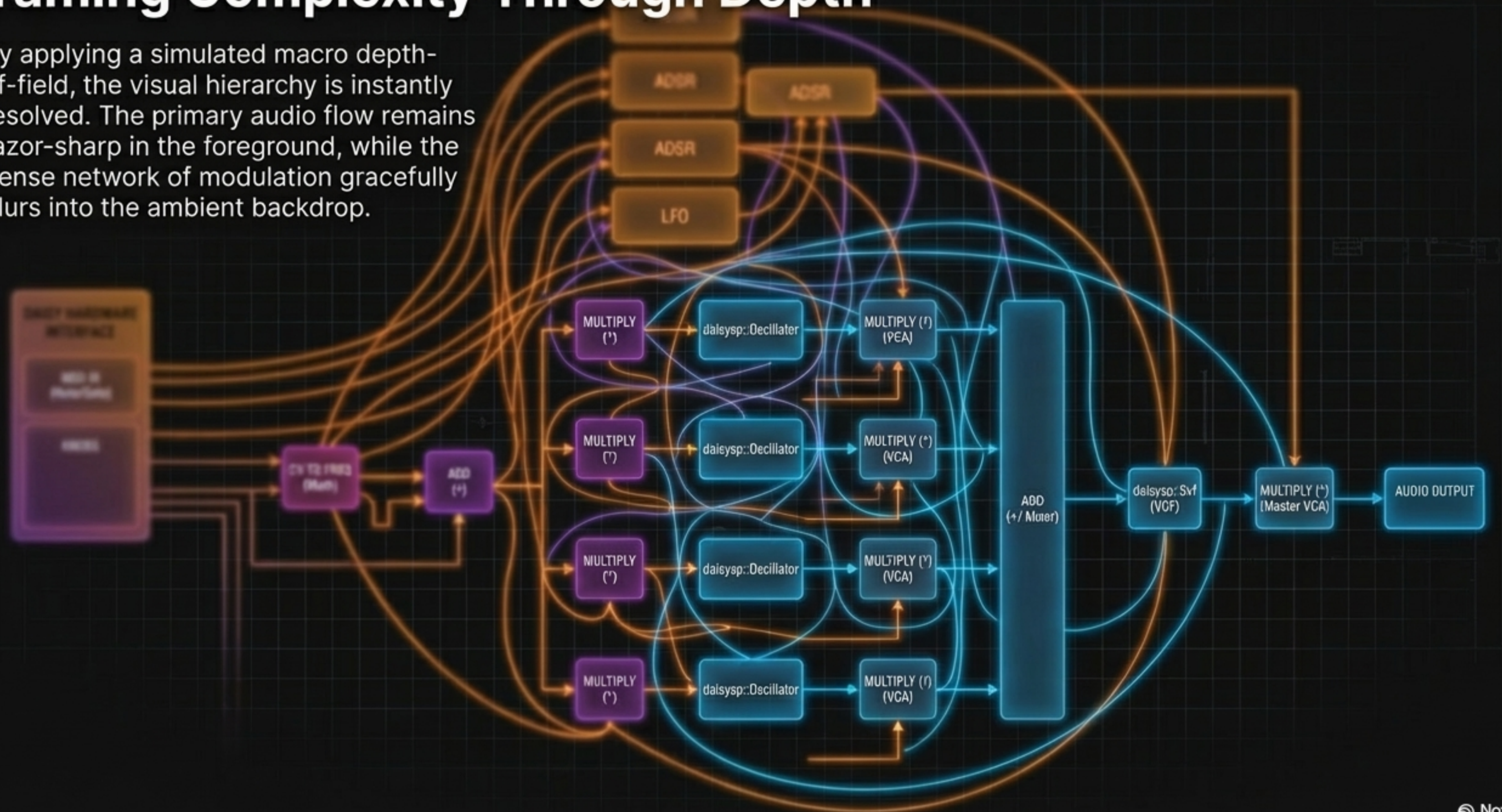
The Density Constraint

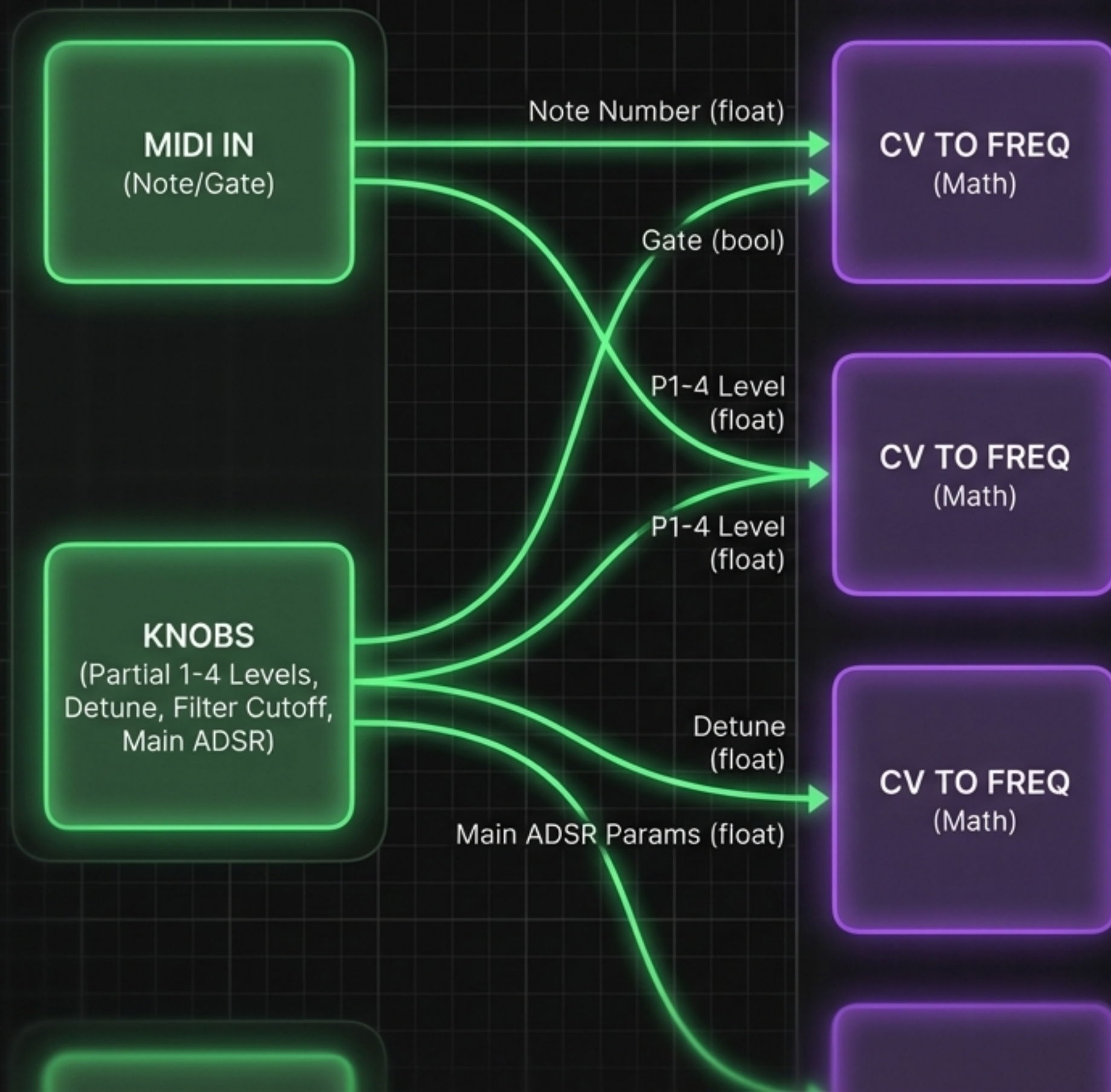
A 4-partial harmonic stack requires 37 independent DSP nodes. Visualizing every base frequency calculation, amplitude envelope, and audio sum on a single flat 2D plane creates immediate visual fatigue.



Taming Complexity Through Depth

By applying a simulated macro depth-of-field, the visual hierarchy is instantly resolved. The primary audio flow remains razor-sharp in the foreground, while the dense network of modulation gracefully blurs into the ambient backdrop.



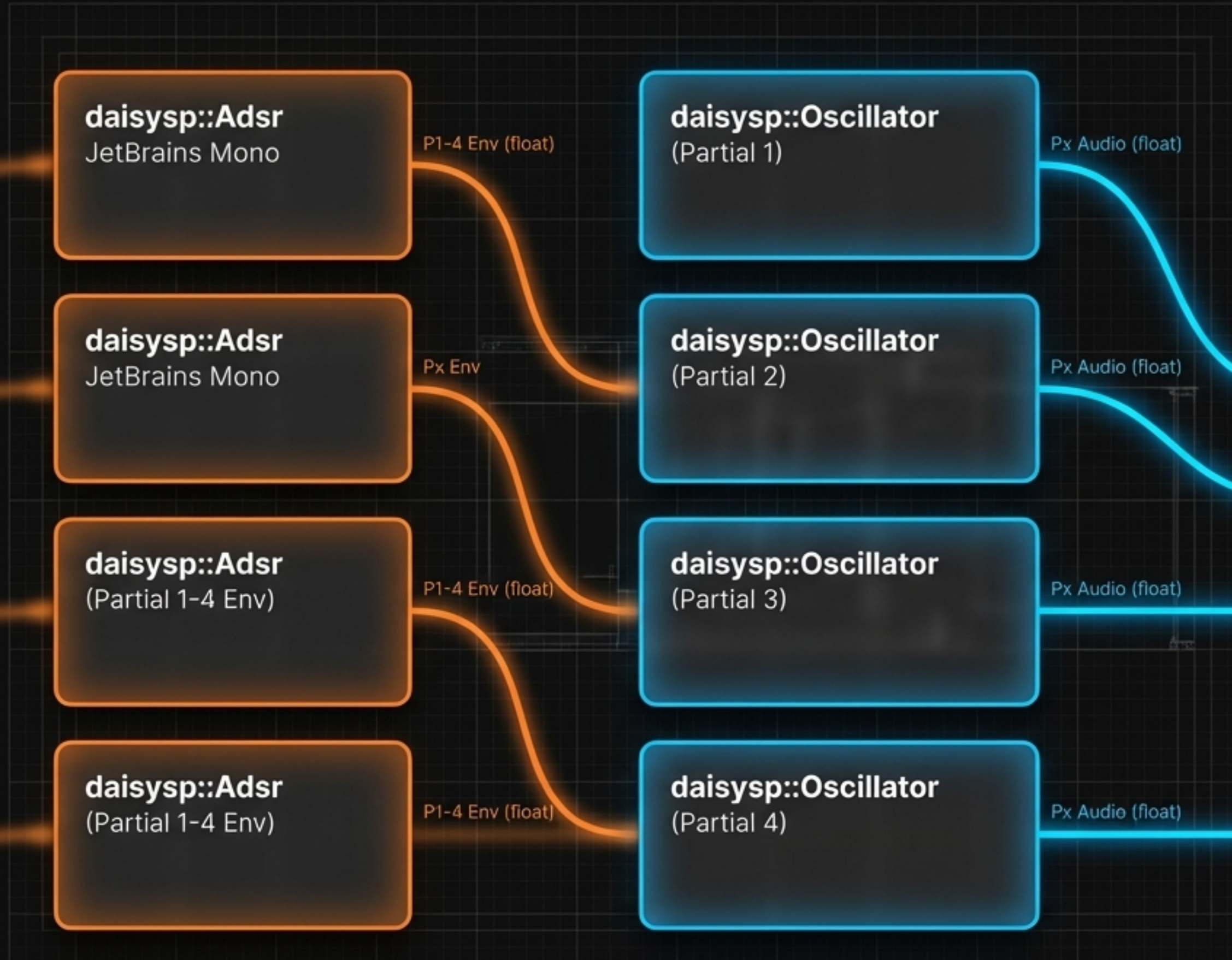


Architecture: The Input Stage

Physical interaction translates into mathematical potential. Hardware knobs and MIDI notes (Green) are converted into base frequency control voltages (Violet) to drive the entire harmonic stack simultaneously.

Architecture: The Harmonic Core

Four distinct oscillators form the additive stack. Each partial is paired with an independent amplitude envelope, allowing precise temporal sculpting of individual harmonics.



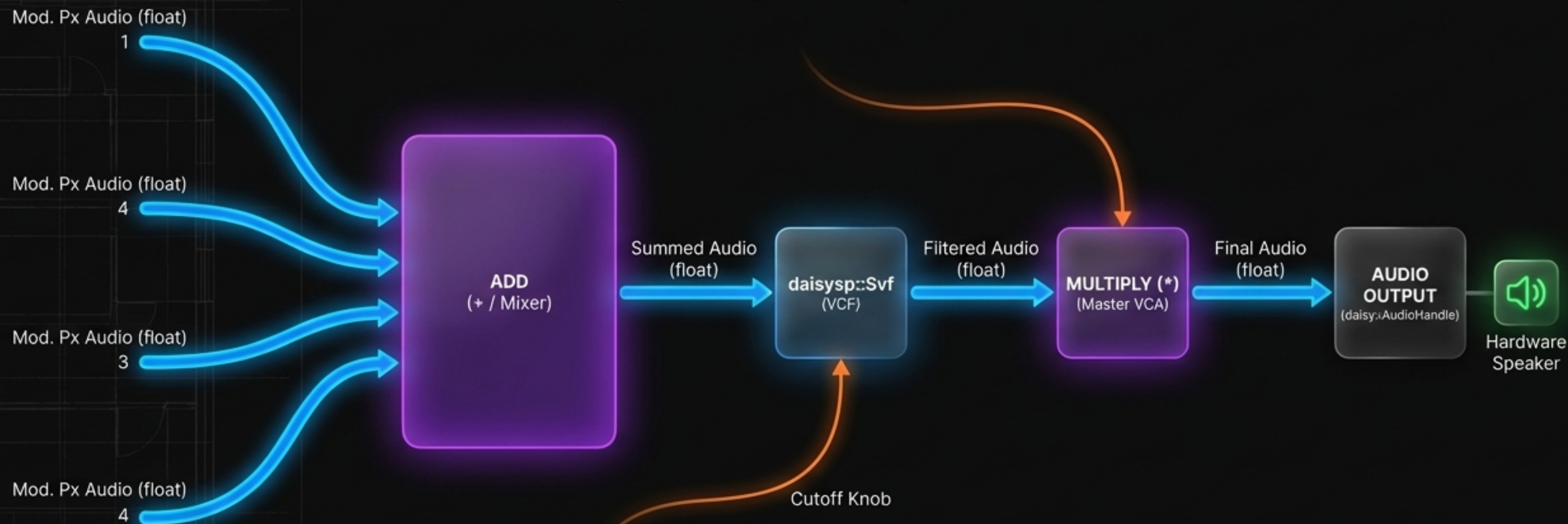


Architecture: Multiplier Logic

The math behind the music. Electric Violet logic nodes calculate the exact pitch of each harmonic partial by multiplying the base fundamental frequency against fixed harmonic constants.

Architecture: The Master Bus

Resolution. The four discrete harmonic partials are summed into a unified waveform, passed through a master voltage-controlled filter, and dynamically shaped by the main amplitude envelope.



Field Additive Synthesizer

Complex digital signal processing rendered as luminous industrial design. 37 nodes. Zero cognitive overload.

